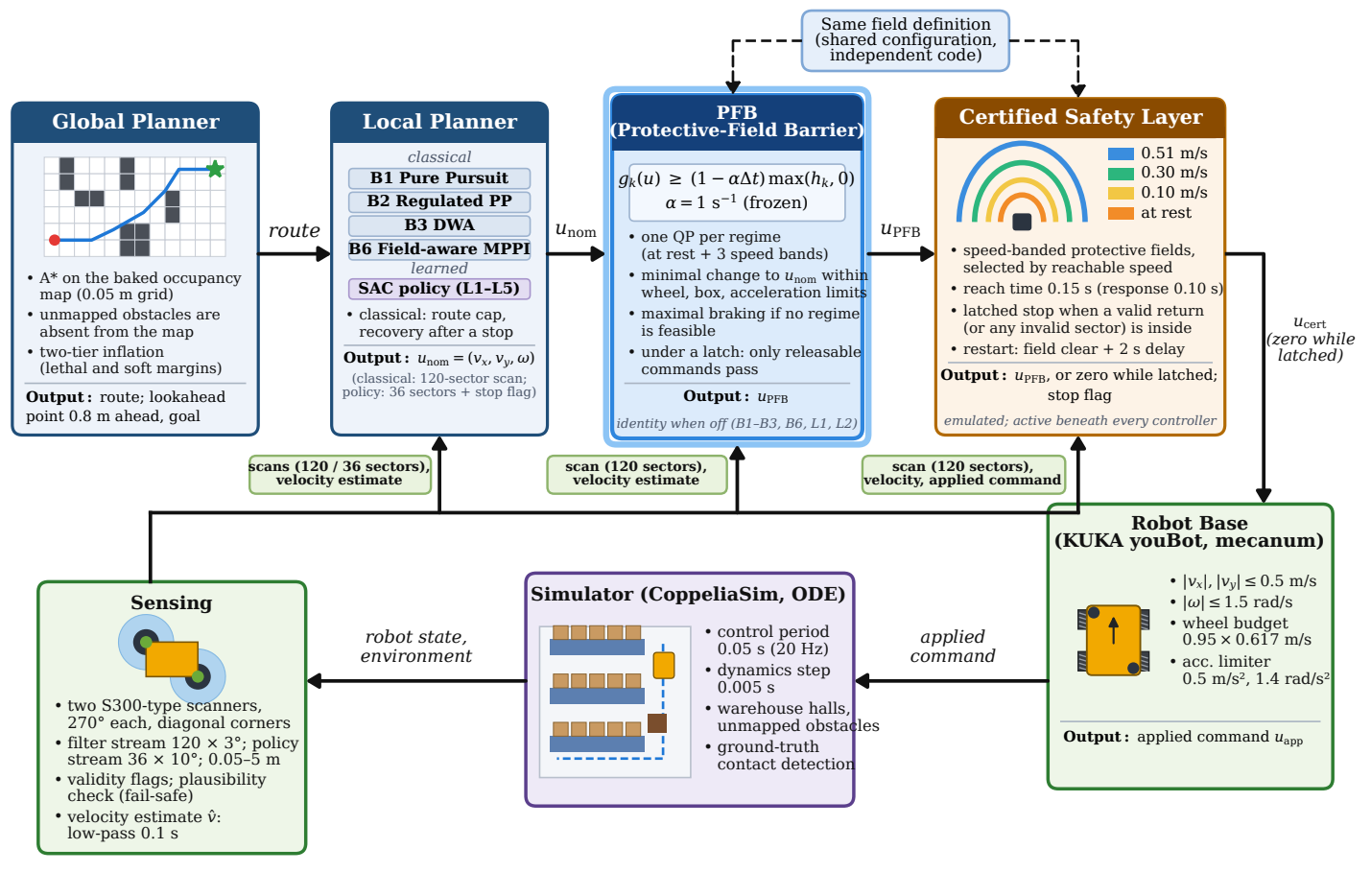
Same field definition
(shared configuration,
independent code)
Global Planner
• A* on the baked occupancy map (0.05 m grid)
• unmapped obstacles are absent from the map
• two-tier inflation (lethal and soft margins)
Output: route; lookahead point 0.8 m ahead, goal
route
Local Planner
classical
B1 Pure Pursuit
B2 Regulated PP
B3 DWA
B6 Field-aware MPPI
learned
SAC policy (L1–L5)
• classical: route cap, recovery after a stop
Output: $u_{nom} = (v_x, v_y, \omega)$
(classical: 120-sector scan; policy: 36 sectors + stop flag)
$u_{nom}$
PFB
(Protective-Field Barrier)
$g_k(u) \geq (1 - \alpha\Delta t)\max(h_k, 0)$
$\alpha = 1\ s^{-1}$ (frozen)
• one QP per regime (at rest + 3 speed bands)
• minimal change to $u_{nom}$ within wheel, box, acceleration limits
• maximal braking if no regime is feasible
• under a latch: only releasable commands pass
Output: $u_{PFB}$
identity when off (B1–B3, B6, L1, L2)
$u_{PFB}$
Certified Safety Layer
0.51 m/s
0.30 m/s
0.10 m/s
at rest
• speed-banded protective fields, selected by reachable speed
• reach time 0.15 s (response 0.10 s)
• latched stop when a valid return (or any invalid sector) is inside
• restart: field clear + 2 s delay
Output: $u_{PFB}$, or zero while latched; stop flag
emulated; active beneath every controller
$u_{cert}$
(zero while latched)
scans (120 / 36 sectors), velocity estimate
scan (120 sectors), velocity estimate
scan (120 sectors), velocity, applied command
Robot Base
(KUKA youBot, mecanum)
• $|v_x|, |v_y| \leq 0.5$ m/s
• $|\omega| \leq 1.5$ rad/s
• wheel budget 0.95 × 0.617 m/s
• acc. limiter 0.5 m/s², 1.4 rad/s²
Output: applied command $u_{app}$
applied command
Simulator (CoppeliaSim, ODE)
• control period 0.05 s (20 Hz)
• dynamics step 0.005 s
• warehouse halls, unmapped obstacles
• ground-truth contact detection
robot state, environment
Sensing
• two S300-type scanners, 270° each, diagonal corners
• filter stream 120 × 3°; policy stream 36 × 10°; 0.05–5 m
• validity flags; plausibility check (fail-safe)
• velocity estimate $\hat{v}$: low-pass 0.1 s